GG-B-686b
AUGUST 13, 1957
SUPERSEDING
Int. Fed. Spec. GG-B-00686a(GSA-FSS)
March 13, 1956 and
Fed. Spec. GG-B-686
October 23, 1945

FEDERAL SPECIFICATION

BROACH, ROOT CANAL, DENTAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers root canal broaches suitable for use in endodontic procedures.

1.2 Classification.

1.2.1 *Types and sizes.*—Broaches shall be furnished in the following types and sizes as specified:

Type I.—Smooth.
Size 2.—Fine.

Type II.—Barbed.
Size 1.—Extra fine.
Size 2.—Fine.
Size 3.—Medium.
Size 4.—Coarse.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

QQ-W-470—Wire, Steel, High Carbon, Spring, Bright, Music.
PPP-B-566 — Boxes, Folding, Paperboard.
PPP-B-585—Boxes, Wood, Wirebound.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-676—Boxes, Set-Up, Paperboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116—Preservation, Methods of.
MIL-B-10377 — Box, Wood, Cleated, Veneer, Paper Overlaid.
MIL-L-10547 — Liners, Case, Waterproof.

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Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Wire.* — Broaches shall be made of bright, music wire conforming to Federal Specification QQ–W–470, except that the tensile strength shall be not less than 220,000 pounds per square inch.

3.2 Description.

3.2.1 *Type I, smooth.* — Smooth broaches shall be suitable for use to explore the spread and direction of the root canal in endodontic procedures.

3.2.2 *Type II, barbed.*—Barbed broaches shall be suitable for use to remove pulp tissue and loose debris, of the root canal in endodontic procedures.

3.2.3 *Shank.*—The shank shall be the full diameter of the wire, 0.030 ± 0.005 inch, and shall be 0.8125 ± 0.0625 inch in length, with the free end rounded or blunted.

3.2.3.1 *Cold bends.*—The shank end of the broaches shall withstand a minimum of eight bends, when tested as specified in 4.3.1.

3.2.3.2 *Recoil.* — The shank end of the broaches shall have a recoil of not less than 22½ degrees, when tested as specified in 4.3.2.

3.2.4 *Working end.* — Working end of broaches shall be uniformly tapered, and the tip shall be bullet-shaped. The barbs shall be peeled or cut, so that 85 percent of the cross section area of the shaft remains. The cut of one barb shall not extend into the cut of another. No barb shall be opposite another. The cross section of each shall increase uniformly from tip to base. Barbs shall neither strip off nor bend out of position, when tested as specified in 4.3.3.

3.3 Dimensions.

3.3.1 *Type I.*—Smooth broaches shall conform to the dimensions specified in table I.

TABLE I.—*Type I, smooth broach*

Size	Diameter 0.50 inch from point (± 0.001)	Diameter 0.003 inch from point (± 0.001)	Length (± 0.062)
	Inch	*Inch*	*Inches*
2	0.01	0.003	1.938

3.3.2 *Type II.*—Barbed broaches shall conform to the dimensions specified in table II.

TABLE II.—*Type II, barbed broaches*

Size	Distance from point to root of first barb (± 0.005)	Diameter 0.063 inch from point (± 0.001)	Diameter over barbs .172 inch from point (± 0.003)	Length of barbed section (± 0.031)	Number of barbs (minimum)	Diameter 0.50 inch from point (± 0.001)	Length of broach (± 0.062)
	Inch	*Inch*	*Inch*	*Inch*		*Inch*	*Inches*
1	0.010	.006	.014	.281	36	.009	1.938
2	0.010	.007	.015	.344	44	.010	1.938
3	0.013	.009	.020	.344	44	.013	1.938
4	0.018	.011	.024	.344	44	.017	1.938

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3.4 **Workmanship.** — The Workmanship shall be first class throughout. The broaches shall be evenly cut, properly polished throughout entire length, and shall be free from all defects which will affect their appearance or impair their serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling.**—Unless otherwise specified, samples shall be submitted in accordance with instructions furnished by the procuring activity (see 6.2). In addition, sampling procedures shall be conducted in accordance with the procedures set forth in Military Standard MIL–STD–105.

4.2 **Inspection.**—Inspection may be made throughout the entire process of manufacture. The passing as satisfactory of any detail of construction or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance. Final inspection of the finished article shall be made either at point of production or at point of delivery as specified, (see 6.2). In case of factory inspection, every facility shall be afforded inspectors, by the manufacturer, for the prosecution of their work.

4.3 **Tests.**

4.3.1 *Cold bend test.*—The bends shall be made on a bending gage at the rate of 90° per second, alternately right and left. Each 90 degrees of travel shall be considered one bend. Trade-marks shall be avoided in this test.

4.3.2 *Recoil test.*—One 90° bend, around a pin ¼-inch in diameter, shall be made on a recoil gage at the rate of 90° per second, and the tension released. The pointer shall be brought up to the bent portion to read the number of degrees of recoil.

4.3.3 *Barb breakage test.*—The shank portion of the barbed broach shall be held by means of hemostatic forceps or other suitable holders. The barbed section shall then be placed between two sheets of smoothly polished acrylic resin, and the required load, as indicated in table III for the respective size, shall be applied to the top surface of the upper sheet. The broach shall then be pulled forward through the two sheets of resin. Only one forward movement shall be considered in this test.

TABLE III.

Size	Load*
1	12 oz.
2	1 lb.
3	2 lbs.
4	4 lbs.

* Applied load includes weight of upper sheet of acrylic resin.

4.3.4 *Measurements.* — The measurements shall be made with a toolmaker's microscope or other instrument of equal accuracy.

4.3.5 *Examination of barb breakage.* — Loss of barbs shall be determined by examination under a microscope.

5. PREPARATION FOR DELIVERY

(For definitions and levels, see 6.1)

5.1 **Packaging.**

5.1.1 *Unit of issue.*—Twelve broaches of one type and class, as specified, constitute one unit.

5.1.2 *Quantities.*—Packaging and packing quantities shall be as specified in table IV.

TABLE IV.—*Packaging and packing quantities*

Unit Pack	Intermediate Pack	Shipping Container
1 unit	12 units	1,440 units

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5.1.2.1 *Packing variation permitted.* — If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.3 *Level A.*

5.1.3.1 *Unit package.*—Each broach shall be thoroughly cleaned and treated with a suitable rust preventive. Twelve broaches (1 unit) shall be packaged in a container of suitable size and design. Broaches shall be secured in such manner as to protect the working edges. Each unit shall be further packaged in accordance with method IA–8 of Military Specification MIL–P–116.

5.1.3.2 *Intermediate package.*—The number of units specified in table IV, shall be packaged in a carton of appropriate size and design constructed in accordance with Federal Specifications PPP–B–566 or PPP–B–676. Carton shall be adequately secured to prevent accidental opening.

5.1.4 *Level B (see section 6).*

5.1.4.1 *Unit package.*—Each broach shall be thoroughly cleaned and treated with a suitable rust preventive. Twelve broaches (1 unit) shall be packaged in a container of suitable size and design. Broaches shall be secured in such manner as to protect the working edges.

5.1.4.2 *Intermediate package.*—The number of units specified in table IV shall be packaged in a carton of appropriate size and design, or shall be completely wrapped with kraft paper of suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

5.1.5 *Level C.*—Unless otherwise specified, commercial packages are acceptable.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Exterior container.* — The number of units specified in table IV, shall be overpacked in an exterior container designed for type 2 load and constructed in accordance with Federal Specification PPP–B–585, style 3, for class 3 use; PPP–B–601, table II, using type III, class 1 plywood; PPP–B–621, table III; or Military Specification MIL–B–10377, table II.

5.2.1.2 *Case liner.*—Each level A box shall be lined with a waterproof case liner conforming to Military Specification MIL–L–10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 *Strapping.* — Strapping (when required) shall be in accordance with applicable box specification.

Note.—Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 *Level B.*

5.2.2.1 *Exterior container.* — The number of units specified in table IV shall be overpacked in an exterior container conforming to Federal Specification PPP–B–601, table I, using class 1 plywood; or an exterior container designed for type 2 load and constructed in accordance with Federal Specification PPP–B–621, table I; PPP–B–585, style 3, for class 2 use; or Military Specification MIL–B–10377, table I.

5.2.3 *Level C.* — The subject commodity shall be packed in substantial commerical containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to the point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 *Unit package.*—Each unit container and bag shall be marked with stock number, item identification, unit, quantity, and name of contractor. In addition, bag used for Level A packaged, shall bear the legend "Packaged Method IA–8".

5.3.2 *Intermediate package and shipping container.*—Each intermediate package and shipping container shall be marked in accordance with Military Standard MIL–STD–129. Service color and service symbol markings are required only on direct shipment from the contractor's plant to overseas activities.

6. NOTES

6.1 Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.2 Ordering data.—Purchasers should exercise any desired options offered herein, and should specify the following:

(a) Title, number, and date of this specification.

(b) Type and size required (see 1.2.1).

(c) Sampling required (see 4.1).

(d) Selection of applicable levels of preservation and packaging, and packing.

(1) Level B packaging is intended to provide economical but limited protection, and should be specified only when it is determined to be appropriate. Effective period of protection is estimated to be twelve months from date of initial packaging.

6.3 Storage surveillance.—Items packaged in accordance with Level B requirements must be inspected to determine condition when not used within the time period indicated in 6.1 (d) (1). Items not used within such time-period must either be repackaged in accordance with Level B requirements in this specification or with Level A requirements if storage beyond an additional year is anticipated.

6.4 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.5 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the title of the specifications or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.6 Transportation description.—Transportation description applicable to this item is:

Dental instruments.

Carload minimum weight 12,000 pounds, subject to Rule 34, Uniform Freight Classification.

Truckload minimum weight 12,000 pounds.

6.7 This specification covers the following items appearing in the Armed Services Medical Stock List:

6520–502–7750 BROACH, ROOT CANAL, Barbed, Coarse, 12s.

6520–502–7850 BROACH, ROOT CANAL, Barbed, Extra Fine, 12s.

6520–502–7950 BROACH, ROOT CANAL, Barbed, Fine, 12s.

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MILITARY INTERESTS:

Army—M
Navy—Md
Air Force.

Copies of this specification may be purchased for .05 cents each.

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